Experiment 8

Student Name: Aaditya Anand **UID: 23BCS10446**
Branch: CSE **Section/Group: KRG 3-A**
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Subject Name: System Design **Subject Code: 23CSH-314**

1. **Aim**: To design a stock trading platform (similar to Zerodha, Groww & Upstox) that enables users to trade stocks, view real-time market data, manage portfolios, and execute buy/sell orders with high consistency and low latency..

2. **Objective**:

- To design a scalable architecture for a stock trading platform.
- To enable real-time stock price updates using WebSockets.
- To support order placement, modification, and cancellation.
- To ensure high consistency and correctness in financial transactions.
- To implement portfolio tracking and watchlist features.

3. **Tools Used:**

- **Frontend (ReactJS, HTML, CSS, JavaScript, WebSocket)**
  o Built responsive UI with real-time stock updates
- • **Backend (Node.js / Java / Spring Boot)**
  o Developed REST APIs for trading, authentication, and portfolio
- • **Database (PostgreSQL / MySQL)**
  o Managed financial transactions and user data
- • **Redis**
  o Used for caching stock prices and improving latency
- • **Message Systems / WebSocket**
  o Enabled real-time updates for stock prices and order status
- • **Deployment (AWS / GCP, Load Balancer, Docker)**
  o Scalable cloud infrastructure with containerization

4. **System Requirements:**

A. Functional Requirements

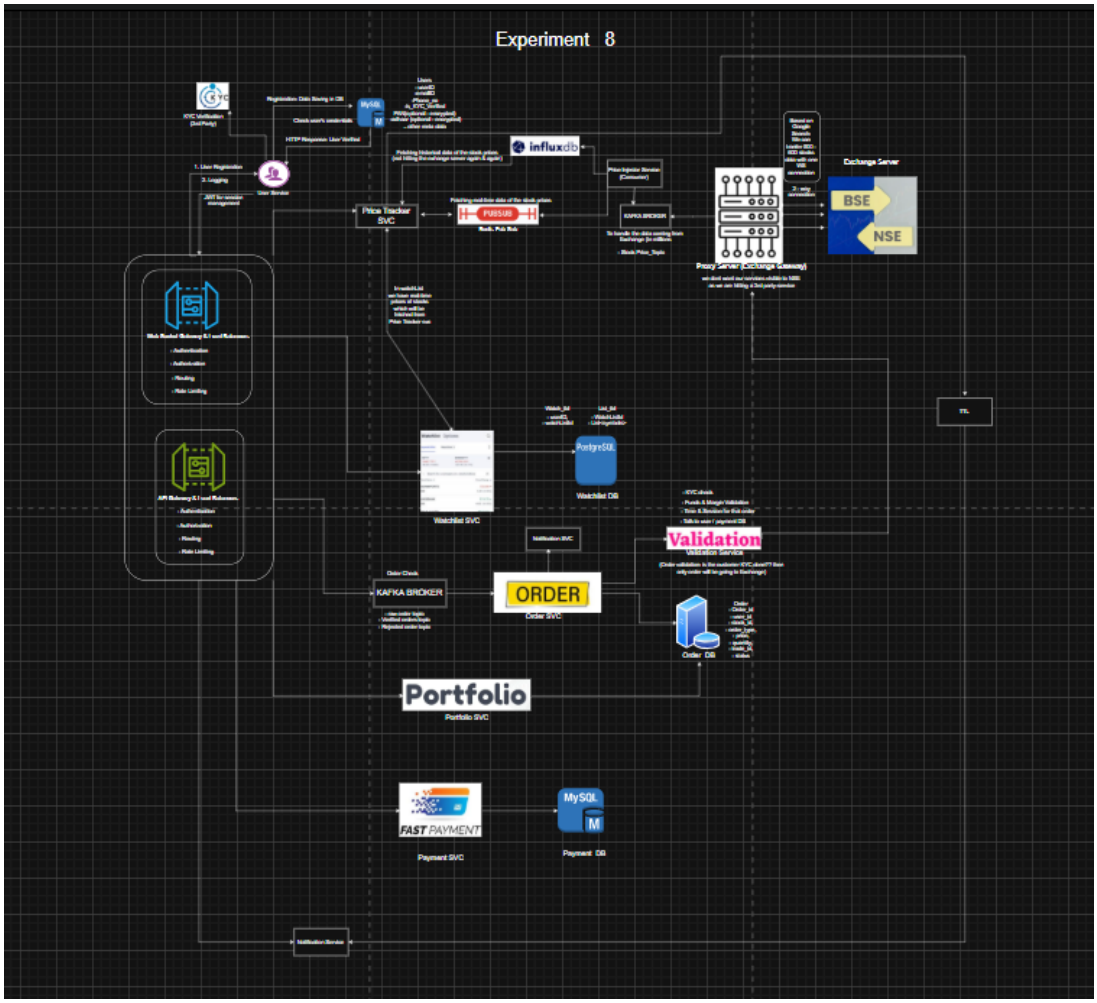
- Users should be able to register, login, and complete KYC verification.
- Users should be able to view real-time stock prices and historical data.
- Users should be able to place, modify, and cancel buy/sell orders.
- The system should support limit orders and market orders.

- Users should be able to maintain a watchlist with real-time updates.
- Users should be able to view portfolio, holdings, and performance.
- The system should provide order status updates in real-time.
- Users should be able to deposit and withdraw funds.

B. Non-Functional Requirements

- **Scalability**
  System should support **2–3 million daily active users** with high-frequency trading and thousands of stocks.
- **CAP Theorem**
  Consistency >>> Availability
  Financial correctness is critical (no duplicate trades, no wrong balances).
- **Latency:**
  Order placement latency should be **< 100 ms**
  Stock price updates latency should be **< 50 ms**
- **Reliability:**
  System must ensure fault tolerance and no data loss in transactions.

5. Low Level Design (LLD):



6. Scalability Solution

- Use **WebSockets** for real-time stock price updates
- Implement **Redis caching** for frequently accessed data
- Use **load balancers** for distributing traffic
- Use **microservices architecture** (Order, Portfolio, Market Data, Funds)
- Store transactional data in **strongly consistent databases**
- Use **horizontal scaling** for handling millions of users
- Optimize read-heavy operations using caching

7. Learning Outcomes (What I Have Learnt)

- Understood architecture of real-time stock trading systems
- Learned importance of **consistency in financial systems**
- Gained knowledge of **WebSocket-based real-time updates**
- Understood order lifecycle (place → match → execute → settle)
- Learned how to design scalable and low-latency systems
- Understood difference between **availability vs correctness trade-offs**.